

Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Introduction & Case's Method

The problem is the monoenergetic, steady-state Boltzmann equation in slab geometry, for an infinite homogeneous isotropic medium with an isotropic plane source at $x = 0$:

$$\mu \frac{\partial \psi(x, \mu)}{\partial x} + \psi(x, \mu) = \frac{c}{2} \int_{-1}^1 \psi(x, \mu') d\mu' + \frac{1}{4\pi} \delta(x) \quad (1)$$

with $c \in [0, 1]$ the scattering ratio and μ the direction cosine. Case (1953) looks for separable solutions $\psi_\nu(x, \mu) = e^{-x/\nu} \phi_\nu(\mu)$, which gives

$$(\nu - \mu) \phi_\nu(\mu) = \frac{c\nu}{2} \int_{-1}^1 \phi_\nu(\mu') d\mu' \xrightarrow{\int \phi_\nu d\mu = 1} (\nu - \mu) \phi_\nu(\mu) = \frac{c\nu}{2} \quad (2)$$

and splits into two spectra:

1. **Discrete** ($\nu \notin [-1, 1]$). Division by $\nu - \mu$ is legal, so $\phi_0^\pm(\mu) = c\nu_0/[2(\nu_0 \mp \mu)]$, and normalization gives the characteristic equation

$$1 = c\nu_0 \operatorname{arctanh}(1/\nu_0) \quad (3)$$

with exactly one pair of real roots $\pm\nu_0$, $\nu_0 > 1$, for $c \in (0, 1)$.

2. **Continuous** ($\nu \in [-1, 1]$). Here $\nu - \mu$ vanishes at $\mu = \nu$, so the solution exists only as a distribution,

$$\phi_\nu(\mu) = \frac{c\nu}{2} \mathcal{P} \frac{1}{\nu - \mu} + \lambda(\nu) \delta(\mu - \nu), \quad \lambda(\nu) = 1 - c\nu \operatorname{arctanh}(\nu) \quad (4)$$

with $\mathcal{P}$ the Cauchy principal value and λ fixed by the same normalization.

The singular objects live in angle only. The scalar flux $\phi(x) = 2\pi \int_{-1}^1 \psi d\mu$ integrates them out, since $\int_{-1}^1 \phi_\nu(\mu) d\mu = 1$ by construction, leaving the ordinary integral

$$\phi_{\text{tr}}(x) = \frac{1}{2} \int_0^1 \frac{e^{-|x|/\nu}}{N_\nu} d\nu, \quad N_\nu = \nu \left[\lambda^2(\nu) + \frac{\pi^2 c^2 \nu^2}{4} \right] \quad (5)$$

whose integrand is smooth on $(0, 1)$: it vanishes at $\nu \rightarrow 0$ through $e^{-|x|/\nu}$ and at $\nu \rightarrow 1$ through $\lambda \rightarrow -\infty$, so the quadrature is well behaved.

2 Question 1: Exact Scalar Flux Analysis

The exact flux is the sum of the two contributions,

$$\phi(x) = \phi_{\text{as}}(x) + \phi_{\text{tr}}(x), \quad \phi_{\text{as}}(x) = \frac{e^{-|x|/\nu_0}}{2N_0^+}, \quad N_0^+ = \frac{c}{2}\nu_0^3 \left[\frac{c}{\nu_0^2 - 1} - \frac{1}{\nu_0^2} \right] \quad (6)$$

ν_0 is computed two ways: by bisection on $\text{arctanh}(y)/y = 1/c$ with $y = 1/\nu_0 \in (0, 1)$, and from the analytic fit

$$\nu_0 \approx \frac{1}{\sqrt{1 - c^{p(c)}}}, \quad p(c) = 2.47412 + \frac{0.00363081}{c^2} - 0.0352458c + \frac{0.557498}{c} \quad (7)$$

which agree to better than $5 \times 10^{-3} \%$ over $c = 0.5 \dots 0.95$. All figures are produced with both.

2.1 Question 1a: Asymptotic and Transient Components

[Insert plots of asymptotic and transient components for each c , and the relative contributions $\phi_{\text{tr}}(x)/\phi(x)$ and $\phi_{\text{as}}(x)/\phi(x)$ as a function of x . Add physical discussion here.]

2.2 Question 1b: Classical Diffusion Solution

In mean free paths ($\Sigma_t = 1$) the diffusion equation for the same source is

$$-D\phi''(x) + (1 - c)\phi(x) = \delta(x) \quad (8)$$

and with the P_1 value $D = 1/3$ its Green's function is

$$\phi_{\text{class}}(x) = \frac{\sqrt{3}}{2\sqrt{1 - c}} e^{-\sqrt{3(1 - c)}|x|} \quad (9)$$

[Insert comparison plots with the classical diffusion solution. Discuss the differences, including the behavior near the source ($x \rightarrow 0$) and the asymptotic behavior ($x \rightarrow \infty$).]

2.3 Question 1c: Asymptotic Diffusion Solution

Matching the diffusion length to the exact eigenvalue, $L = \nu_0$, fixes $D_{\text{as}} = (1 - c)\nu_0^2$ and gives

$$\phi_{\text{as,diff}}(x) = \frac{1}{2(1 - c)\nu_0} e^{-|x|/\nu_0} \quad (10)$$

which is exactly Case's asymptotic mode: this approximation reproduces ϕ_{as} by construction and errs only by the transient part.

[Insert comparison plots with the asymptotic diffusion solution. Discuss how the asymptotic diffusion solution compares to both the classical diffusion solution and the exact transport asymptotic component $\phi_{\text{as}}(x)$.]

2.4 Question 1d: Relative Error of Diffusion Approximations

$$\text{Relative Error (\%)} = \frac{|\phi_{\text{approx}}(x) - \phi_{\text{exact}}(x)|}{\phi_{\text{exact}}(x)} \times 100\% \quad (11)$$

[Insert relative error plots as a function of x . Add physical discussion comparing the accuracy of the classical and asymptotic diffusion approximations across the different values of c . Analyze why the error behaves as it does near the source and far from it.]

Exact transport vs. diffusion (numerical)

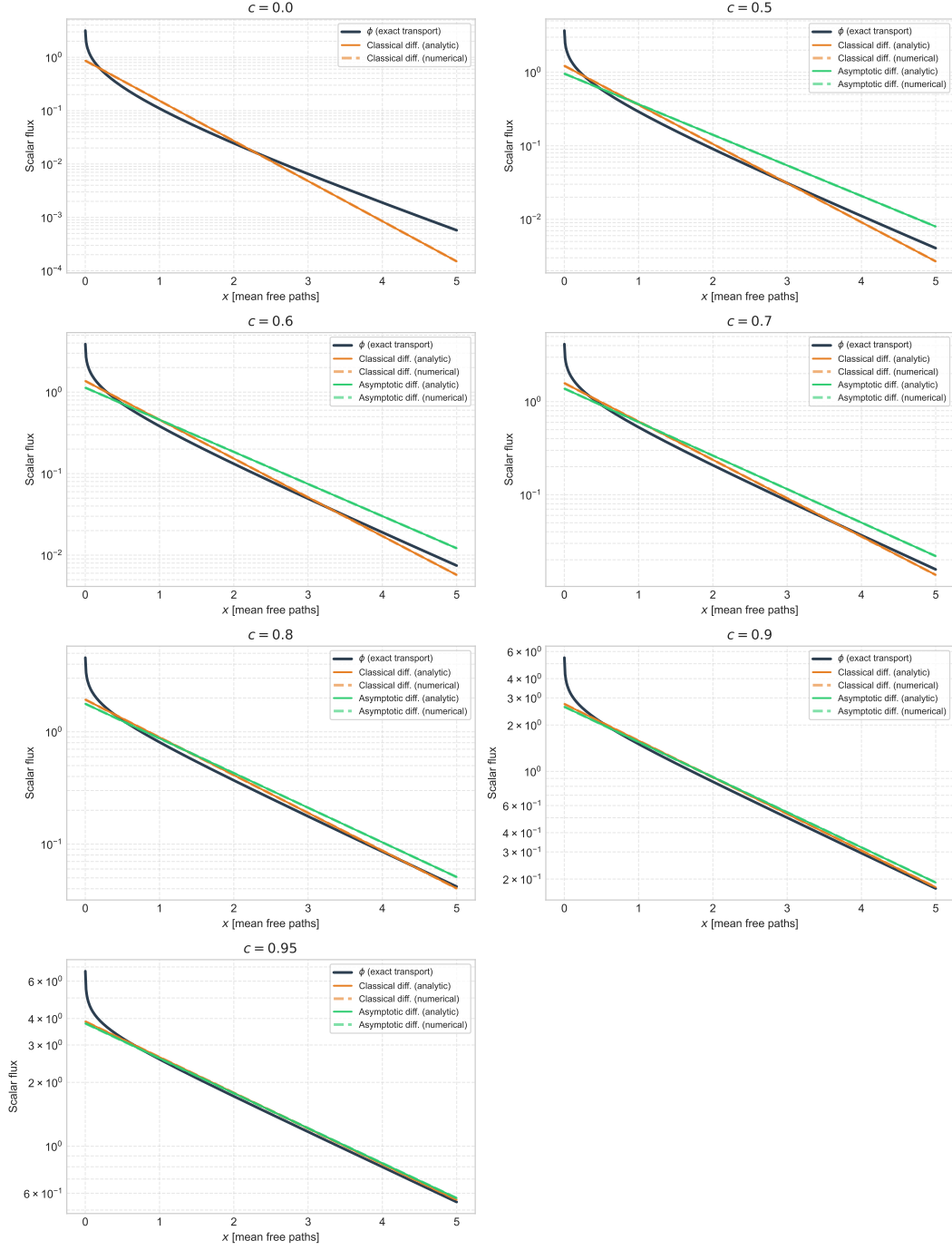


Figure 1: Exact scalar flux against the classical and asymptotic diffusion approximations, for $c = 0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95$. Each approximation appears both as its closed form and as the numerical solution of Question 2; the two agree to $2 \times 10^{-8} \%$ and are indistinguishable here.

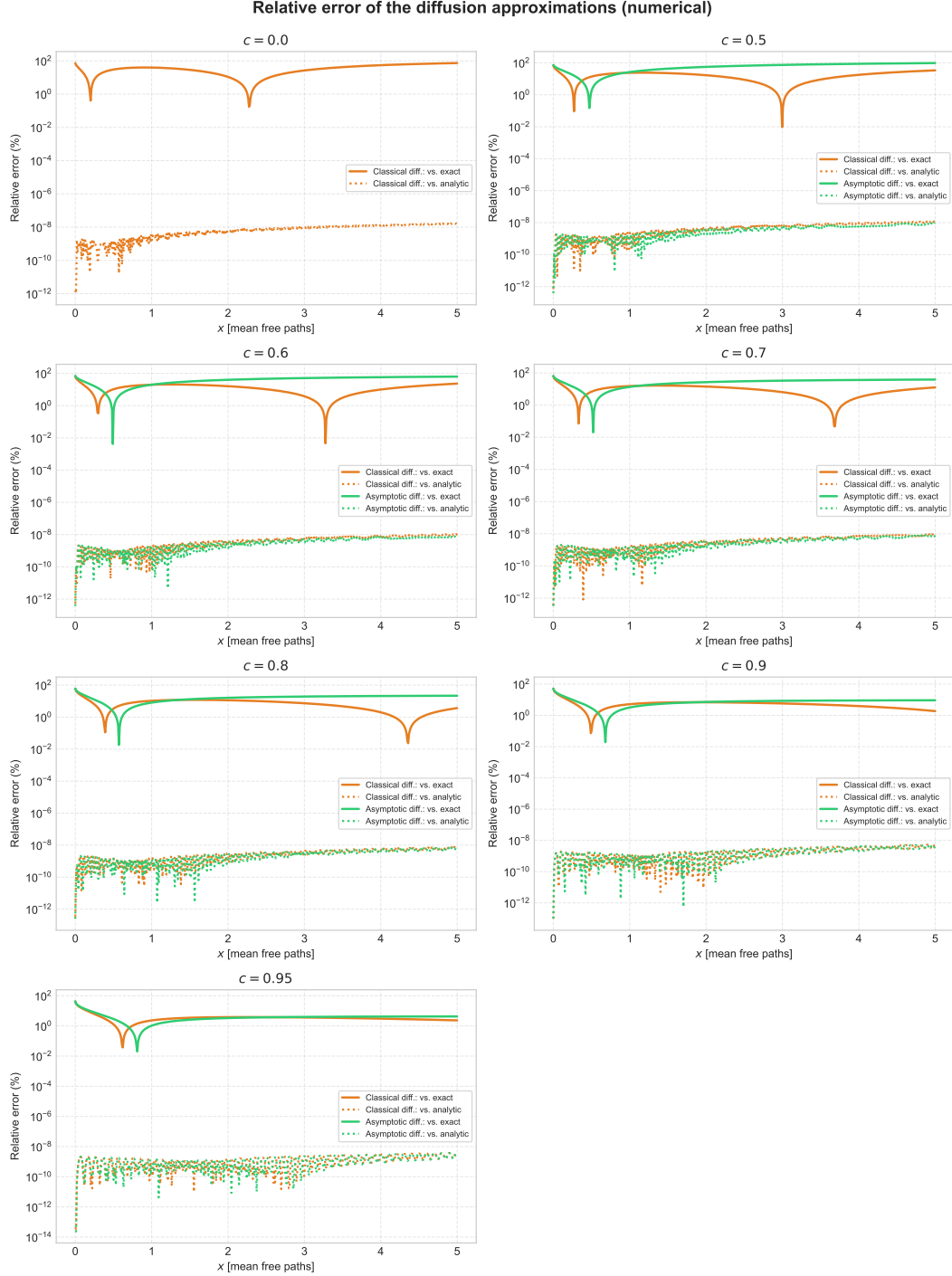


Figure 2: Two errors per approximation: against the *exact transport* solution (solid), which is the modelling error asked for in 1d, and against the *analytic diffusion* solution (dotted), which is the solver error asked for in Question 2. They differ by about ten orders of magnitude, so the solid curves may be read as properties of the approximation rather than of the code.

3 Question 2: Numerical Code for the Diffusion Equation

The medium is symmetric, so the problem is solved on the half-domain $x \in [0, a]$, where it is source-free:

$$-D\phi''(x) + (1 - c)\phi(x) = 0 \quad \Rightarrow \quad \frac{d^4}{dx} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ \frac{1-c}{D} y_1 \end{pmatrix}, \quad y_1 = \phi, \quad y_2 = \phi' \quad (12)$$

The delta source. It is never discretised. Integrating across the origin gives $-D[\phi'(0^+) - \phi'(0^-)] = 1$, and evenness of the infinite-medium solution gives

$$\phi'(0^+) = -\frac{1}{2D}, \quad \text{i.e.} \quad J(0^+) = -D\phi'(0^+) = \frac{1}{2} \quad (13)$$

half the source streaming each way. That is how a delta source should be modelled: symmetry converts it into a prescribed current and the singularity never enters the scheme.

The outer boundary. $x = a$ is a truncation, not a surface. Zero flux there would force a 100% error at $x = a$ that no tolerance removes, so the radiation condition

$$\phi'(a) = -\kappa\phi(a), \quad \kappa = \sqrt{\Sigma_a/D} \quad (14)$$

is used instead: $e^{-\kappa x}$ satisfies it identically, the growing mode is annihilated, and the truncation is exact for any a . We take $a = 10/\kappa$.

One integration. The equation is linear and homogeneous away from the source, so the solution is proportional to its starting amplitude and no root-finding is needed. We integrate once from a to 0 — the direction in which the solution grows, which is the stable one — from $[y_1, y_2]^T = [1, -\kappa]^T$, then rescale by $[-1/(2D)]/y_2(0)$ to enforce the source condition. Both approximations use this one solver, since

$$\Sigma_a = 1 - c, \quad D_{\text{classical}} = \frac{1}{3}, \quad D_{\text{asymptotic}} = (1 - c)\nu_0^2 \quad (15)$$

Verification. The solver reproduces the closed form to $2 \times 10^{-8}\%$ for both approximations at $c = 0.5, 0.7, 0.9$ (Figures 1–4), and conserves neutrons to $\Sigma_a \int \phi dx = 0.99998807$, the deficit being the physical tail beyond the truncation. A shooting method has no mesh to refine, so convergence is shown against the integrator tolerance: the relative L_2 error falls from 9×10^{-6} at $\text{rtol} = 10^{-4}$ to 1.3×10^{-12} at 10^{-11} , with unit slope (Figure 5).

4 Question 3: Critical Half-Thickness and Critical Radius

Parts 3(a)–3(c) are covered: the exact-transport relations, and classical diffusion with the Marshak and Mark conditions. Parts 3(d)–3(e) are not.

4.1 The Eigenvalue for $c > 1$

The discrete eigenvalue leaves the real axis. With $\nu_0 = i|\nu_0|$, $k_0 = 1/|\nu_0|$ and $\text{arctanh}(ik) = i \arctan(k)$, the characteristic equation becomes real again,

$$c \arctan(k_0) = k_0 \quad (16)$$

the form tabulated by Case, de Hoffmann & Placzek (Table 8, Part II), which the root reproduces to 5×10^{-6} . The fit of Question 1 continues onto this branch by itself, its radicand simply changing sign:

$$|\nu_0(c)| = \frac{1}{\sqrt{c^{p(c)} - 1}} \quad (17)$$

Question 2: numerical solver vs. analytic Green's function

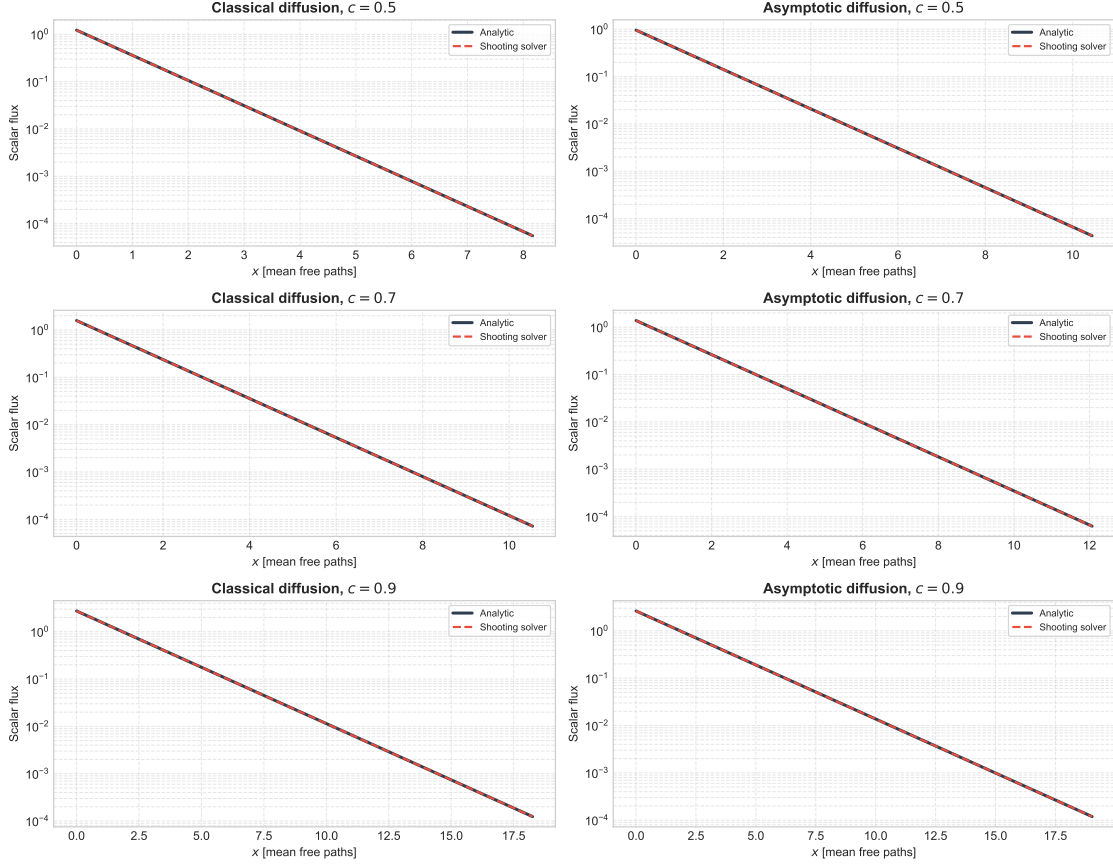


Figure 3: Numerical solution against the analytic diffusion Green's function, both approximations at $c = 0.5, 0.7, 0.9$.

accurate to $10^{-4}\%$ at $c = 1.02$ and 0.10% at $c = 2$. An imaginary ν_0 makes the asymptotic flux trigonometric, which is what allows a finite critical system:

$$\phi_{\text{as}}(x) = A \cos\left(\frac{\Sigma_t x}{|\nu_0|}\right), \quad \phi_{\text{as}}(r) = A \frac{\sin(\Sigma_t r / |\nu_0|)}{\Sigma_t r} \quad (18)$$

the sine being excluded in the slab by symmetry, the cosine in the sphere by regularity.

4.2 The Criticality Relations

The asymptotic flux extrapolates to zero a distance $z_0(c)$ outside the surface. Placing the quarter-wave zero of the cosine and the first zero of the sine there gives, in mean free paths,

$$\frac{a}{2} = \frac{\pi}{2} |\nu_0(c)| - z_0(c), \quad \Sigma_t R_c = \pi |\nu_0(c)| - z_0(c) \quad (19)$$

No diffusion coefficient enters, so these are exact transport in that sense; they are still asymptotic, in that the transient flux near the surface is neglected (and the sphere uses the plane Milne z_0 , without the Winslow curvature correction). The extrapolation distance comes from the expansion about $c = 1$,

$$z_0(c) = 0.710446 \left[\frac{1}{c} + q \frac{(1-c)^2}{c} + \mathcal{O}\left(\frac{(1-c)^3}{c}\right) \right] \quad (20)$$

Question 2: relative error of the numerical solver

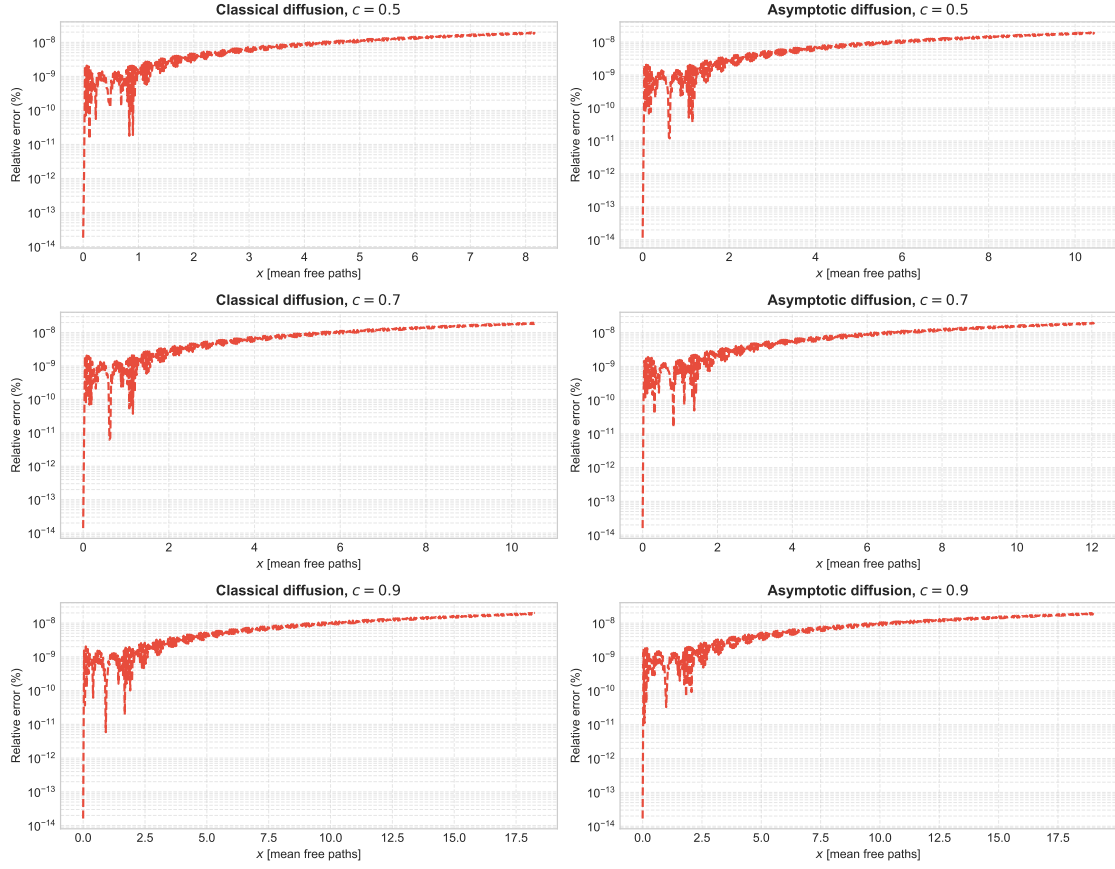


Figure 4: Relative error across the domain. With the radiation condition it stays flat instead of diverging at the outer edge.

Question 2: tolerance convergence of the shooting solver

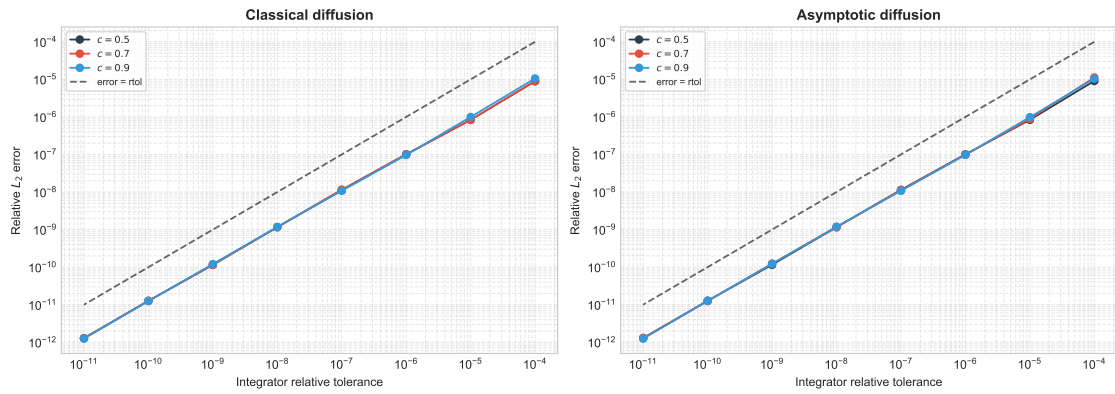


Figure 5: Tolerance convergence of the shooting solver.

whose leading term is exact in the sense that $0.710446/c = 0.710446 \nu_0 \operatorname{arctanh}(1/\nu_0)$, with 0.710446 Hopf's constant.

4.3 The Sign of the Quadratic Correction

The notes print $q = -0.0199$, but Case's Table 23 shows $c z_0(c)$ rising away from 0.710446 on *both* sides of $c = 1$ (Figure 7), which a negative q cannot reproduce; with $q = +0.0199$ the fit matches the table to its four printed digits at $c = 0.9$ and $c = 1.1$. The printed minus sign therefore looks like a transcription error. Over $1.02 < c < 2$ the two signs differ by at most 2.6 % in z_0 and 3.2 % in $a/2$, and by less than 0.07 % in $a/2$ below $c = 1.2$. Results are quoted with the printed sign; both are plotted.

4.4 Results

c	$ \nu_0(c) $	$z_0(c)$	$a/2$	$\Sigma_t R_c$	error in $a/2$
1.02	4.050190	0.696510	5.665513	12.027537	0.0001 %
1.05	2.531781	0.676582	3.300331	7.277244	0.0025 %
1.10	1.756653	0.645731	2.113612	4.872955	0.0128 %
1.20	1.198272	0.591567	1.290674	3.172916	0.0667 %
1.30	0.946021	0.545518	0.940488	2.426495	0.1777 %
1.40	0.793813	0.505846	0.741072	1.987990	0.3592 %
1.50	0.689210	0.471274	0.611333	1.693941	0.6116 %
1.60	0.611746	0.440848	0.520080	1.481008	0.9535 %
1.70	0.551520	0.413834	0.452491	1.318817	1.3672 %
1.80	0.503065	0.389665	0.400547	1.190759	1.8831 %
1.90	0.463074	0.367892	0.359503	1.086897	2.4781 %
2.00	0.429411	0.348154	0.326364	1.000881	3.1859 %

Table 1: Critical dimensions in mean free paths, from the approximate $|\nu_0(c)|$ and the printed z_0 fit ($q = -0.0199$). The last column is the departure of $a/2$ from a reference using the transcendental root and Case's Table 23.

Both dimensions fall steeply with c , from $a/2 = 5.67$, $\Sigma_t R_c = 12.03$ at $c = 1.02$ to 0.33 and 1.00 at $c = 2$: a barely multiplying system must be many mean free paths across, because $|\nu_0| \rightarrow \infty$ as $c \rightarrow 1^+$.

The error of the approximate route is dominated by z_0 , not by $|\nu_0|$ — the eigenvalue fit stays under 0.11 %, while the two-term z_0 expansion reaches 2.6 % by $c = 2$, where $(1 - c)^3$ is no smaller than $(1 - c)^2$. Since z_0 is subtracted while $|\nu_0|$ is multiplied by π or $\pi/2$, the same absolute z_0 error hurts $a/2$ about twice as much as R_c , and hurts both more as the system shrinks: under 0.1 % up to $c \approx 1.25$, a few percent by $c = 2$ (Figure 6).

4.5 Parts 3(b) and 3(c): Marshak and Mark Conditions

A multiplying medium has $\Sigma_a = 1 - c < 0$, so classical diffusion gives a Helmholtz equation

$$-D\nabla^2\phi + (1 - c)\phi = 0 \implies \nabla^2\phi + B^2\phi = 0, \quad B^2 = \frac{c - 1}{D} = 3(c - 1) \quad (21)$$

whose solutions are the *same* shapes $\cos(Bx)$ and $\sin(Br)/r$. Diffusion therefore leaves (19) unchanged in form and only replaces the two inputs:

$$|\nu_0(c)| \longrightarrow \frac{1}{B} = \frac{1}{\sqrt{3(c - 1)}}, \quad z_0(c) \longrightarrow \begin{cases} 2/3 & \text{Marshak} \\ 1/\sqrt{3} & \text{Mark} \end{cases} \quad (22)$$

Question 3: critical dimensions from the transport relations

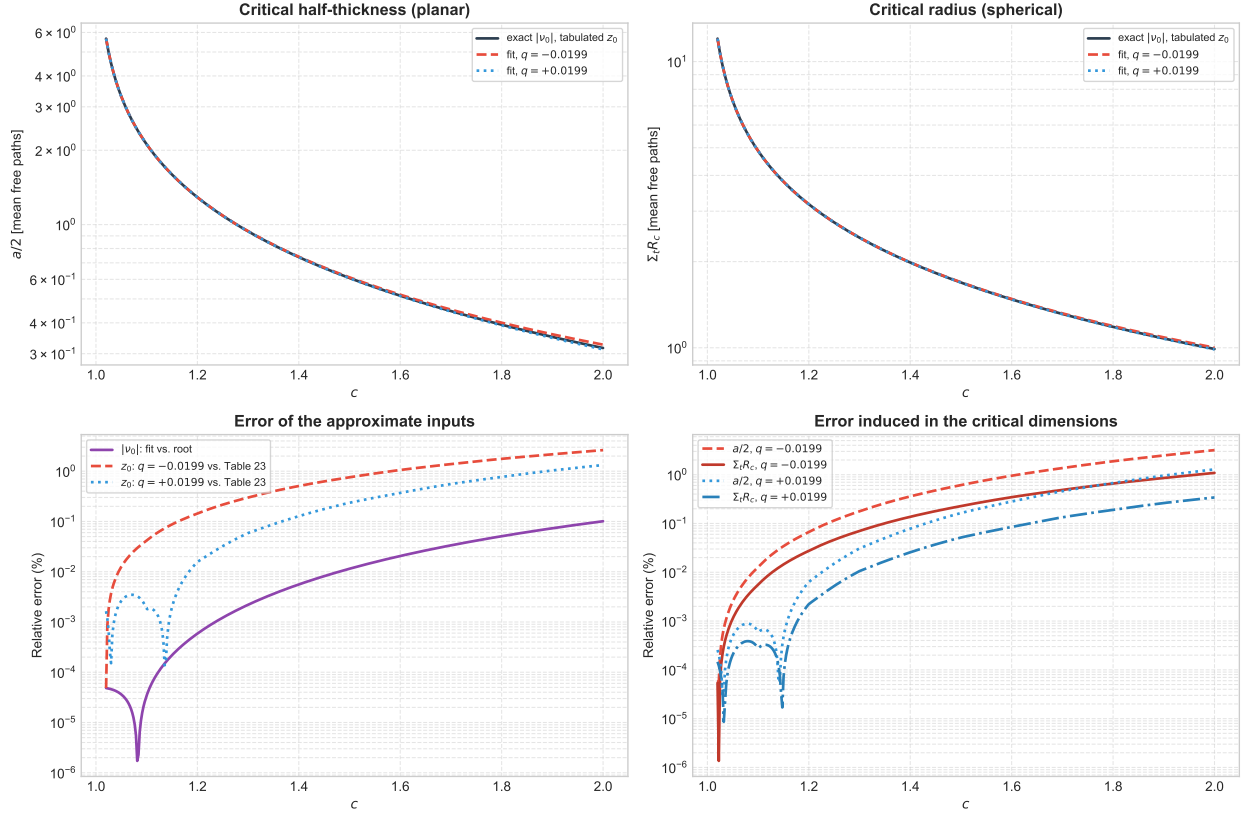


Figure 6: Critical half-thickness and radius for $1.02 < c < 2$ (top), and the errors of the approximate inputs and of the resulting dimensions (bottom).

Marshak sets the incoming partial current to zero, $\phi/4 + D\phi'/2 = 0$, so $z_0 = \ell_0 = 2D = 2/3$; Mark sets the incoming angular flux to zero along $\mu = 1/\sqrt{3}$, so $z_0 = \ell_0 = \sqrt{3}D \approx 0.5774$. Neither input carries the true c -dependence: $1/B$ is only the $c \rightarrow 1^+$ limit of $|\nu_0|$, and both z_0 are constants while the transport $z_0(c) \approx 0.7104/c$ halves across the range.

4.6 Comparison

All three agree as $c \rightarrow 1^+$, where the system is many mean free paths across and the boundary layer is a small correction, and separate as it shrinks (Figure 8). The two diffusion errors act in opposite directions:

- $1/B$ over-estimates the relaxation length — by 0.8 % at $c = 1.02$ and 34 % at $c = 2$ — inflating both dimensions;
- the constant z_0 over-estimates the extrapolation distance as c grows and, being subtracted, deflates both.

The relaxation length is weighted by π in the sphere but only $\pi/2$ in the slab, while z_0 enters both identically. So the spherical radius is over-estimated throughout, by +15 % (Marshak) and +24 % (Mark) at $c = 2$, whereas Marshak's $a/2$ is non-monotonic: +5.5 % at $c \approx 1.23$, zero near $c = 1.52$,

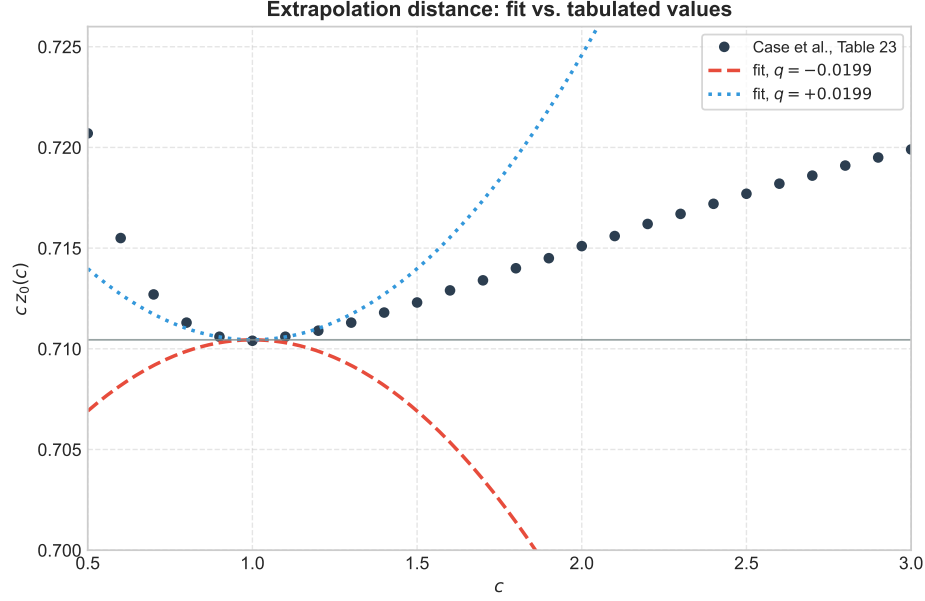


Figure 7: The tabulated product $cz_0(c)$ against the fit with either sign of q . The tabulated values rise on both sides of $c = 1$, which fixes the sign as positive.

c	$a/2$			$\Sigma_t R_c$		
	transport	Marshak	Mark	transport	Marshak	Mark
1.02	5.665513	5.746082	5.835399	12.027537	12.158832	12.248148
1.05	3.300331	3.389112	3.478428	7.277244	7.444891	7.534207
1.10	2.113612	2.201202	2.290518	4.872955	5.069071	5.158387
1.20	1.290674	1.361223	1.450539	3.172916	3.389112	3.478428
1.30	0.940488	0.989098	1.078414	2.426495	2.644863	2.734179
1.40	0.741072	0.767268	0.856584	1.987990	2.201202	2.290518
1.50	0.611333	0.615883	0.705200	1.693941	1.898433	1.987749
1.60	0.520080	0.504136	0.593452	1.481008	1.674938	1.764255
1.70	0.452491	0.417286	0.506602	1.318817	1.501238	1.590555
1.80	0.400547	0.347278	0.436594	1.190759	1.361223	1.450539
1.90	0.359503	0.289290	0.378606	1.086897	1.245246	1.334562
2.00	0.326364	0.240233	0.329549	1.000881	1.147133	1.236449

Table 2: Critical dimensions in mean free paths from the three treatments: exact transport (3a), Marshak diffusion (3b) and Mark diffusion (3c).

−26 % at $c = 2$. Mark’s slab result ends at only +1 % at $c = 2$, which is cancellation rather than accuracy.

That reading is confirmed by dropping the shortcut $z_0 = \ell_0$ and imposing $\phi + \ell_0 \phi' = 0$ on the flux shape itself,

$$B \frac{a}{2} = \arctan\left(\frac{1}{B\ell_0}\right), \quad u \cot u = 1 - \frac{u}{B\ell_0}, \quad u = BR \in (0, \pi) \quad (23)$$

against $Ba/2 = \pi/2 - B\ell_0$ and $u = \pi - B\ell_0$ for the extrapolated form. Since $\arctan(1/y) =$

$\pi/2 - y + \mathcal{O}(y^3)$ the two agree to first order and coincide as $c \rightarrow 1$ (0.1% in the slab at $c = 1.02$), but part company as the system shrinks: at $c = 2$ the Marshak slab gives $a/2 = 0.412$ applied against 0.240 extrapolated. The applied-condition results lie *above* the transport answer in every case.

Question 3: exact transport vs. Marshak and Mark diffusion

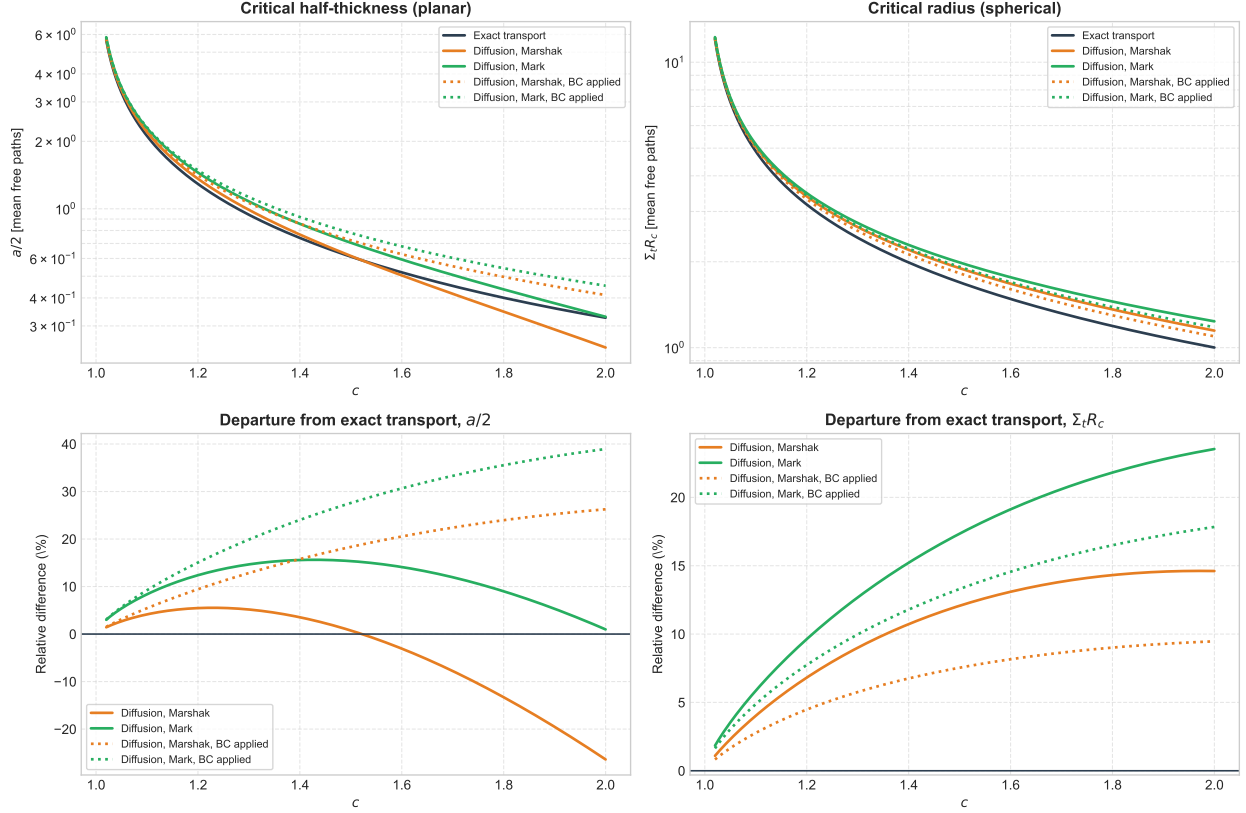


Figure 8: Exact transport against Marshak and Mark diffusion (top), and the signed departure of each from transport (bottom). Dotted curves impose the boundary condition on the flux shape instead of using an extrapolated zero.

5 Question 4: Numerical Diffusion Code in Spherical Coordinates

The code solves

$$-D \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) + \Sigma_a \phi(r) = \frac{1}{k} \nu \Sigma_f \phi(r) \quad (24)$$

and finds the radius at which $k = 1$. Dividing the fission source by k makes a solution exist at every radius, so criticality is a root search on R rather than an eigenvalue search. Both approximations use the same solver, differing in two constants:

$$\text{classical: } D = \frac{1}{3\Sigma_t}, \quad z_0 = 2D, \quad \text{asymptotic: } D = \frac{(c-1)|\nu_0(c)|^2}{\Sigma_t}, \quad z_0 = z_0(c) \quad (25)$$

The asymptotic D is the continuation of Question 2's $(1-c)\nu_0^2$ above $c = 1$: both ν_0^2 and $1-c$ change sign, so D stays positive. Its purpose is to give the diffusion equation the true transport relaxation length $|\nu_0|/\Sigma_t$ instead of $1/\sqrt{3(c-1)}$.

5.1 Discretisation

Finite volume on N uniform cell-centred shells. Integrating (24) over cell i turns the divergence into a difference of surface currents,

$$A_{i+1/2}J_{i+1/2} - A_{i-1/2}J_{i-1/2} + \Sigma_a V_i \phi_i = S_i V_i, \quad J_{i+1/2} = -D \frac{\phi_{i+1} - \phi_i}{h} \quad (26)$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r_{i+1/2}^3 - r_{i-1/2}^3)$. The balance form is what removes the $1/r^2$ singularity: the innermost face has $A_{1/2} = 0$, so $\phi'(0) = 0$ is imposed exactly with no special case. The matrix is tridiagonal and solved in $O(N)$.

Both outer treatments are the single condition $\phi + \ell_0 \phi' = 0$; eliminating the surface flux gives one leakage coefficient

$$J_s = \frac{D \phi_{N-1}}{\ell_0 + h/2} \quad (27)$$

covering the extrapolated zero ($\ell_0 = 0$ on a mesh run to $R + z_0$) and the Robin condition ($\ell_0 = z_0$ at R) without a branch. The extrapolated form is used, since it is what the Question 3 relations assume, making the comparison a test of the numerics alone.

5.2 The Iterative Algorithm

k comes from the source iteration of Bell & Glasstone, pp. 189–192: start from a flat flux and $k = 1$; form $S = \frac{1}{k} \nu \Sigma_f \phi$; solve the removal problem $(-D\nabla^2 + \Sigma_a)\phi^{(n+1)} = S$; update

$$k^{(n+1)} = k^{(n)} \frac{\int \nu \Sigma_f \phi^{(n+1)} dV}{\int \nu \Sigma_f \phi^{(n)} dV} \quad (28)$$

and repeat until $|k^{(n+1)} - k^{(n)}| < 10^{-10} k^{(n+1)}$. The flux is renormalised each sweep: its amplitude grows by roughly k per sweep, and over several hundred sweeps the convergence test would otherwise lose its significant digits. This is physically irrelevant, since the ratio is homogeneous in ϕ .

Convergence is at the dominance ratio k_1/k_0 , which tends to unity for a large weakly multiplying sphere (≈ 0.94 at $c = 1.02$) and is derived in §5.4. The critical radius then follows from Brent bisection on $k(R) - 1$, bracketed within $\pm 5\%$ of the analytic radius so that k is never evaluated far from criticality.

5.3 Comparison with the Analytical Result

At $k = 1$ equation (24) is Helmholtz with $B^2 = (\nu \Sigma_f - \Sigma_a)/D = \Sigma_t(c - 1)/D$, and its regular solution $\sin(Br)/r$ vanishes at the extrapolated radius when

$$R_c = \frac{\pi}{B} - z_0 \quad (29)$$

Substituting the two parameter sets recovers the Question 3 relations exactly: $\Sigma_t R_c = \pi/\sqrt{3(c-1)} - 2/3$ classically, and $\Sigma_t R_c = \pi|\nu_0| - z_0(c)$ asymptotically, the latter being part 3(a). Note that R_c depends on the cross sections only through c , since Σ_a and $\nu \Sigma_f$ enter only as their difference.

The residual is discretisation error alone, and the scheme is second order: at $c = 1.5$ the relative error of R_c falls from 5.28×10^{-4} at $N = 25$ to 5.16×10^{-7} at $N = 800$, a factor of exactly 4.00 per doubling (Figure 10). Its sign is consistently negative — the discrete operator is slightly too leaky on a coarse mesh, so it reaches $k = 1$ at a marginally smaller radius. Two further checks: k at

Table 3: Critical radius in mean free paths from the $k = 1$ search ($N = 400$) against (29).

c	Classical		Asymptotic		difference (classical)
	numerical	analytic	numerical	analytic	
1.02	12.15881	12.15883	12.02752	12.02754	$-1.6 \times 10^{-4} \%$
1.10	5.06906	5.06907	4.87295	4.87295	$-1.7 \times 10^{-4} \%$
1.20	3.38911	3.38911	3.17289	3.17289	$-1.8 \times 10^{-4} \%$
1.50	1.89843	1.89843	1.69369	1.69369	$-2.1 \times 10^{-4} \%$
1.80	1.36122	1.36122	1.18995	1.18996	$-2.3 \times 10^{-4} \%$
2.00	1.14713	1.14713	0.99952	0.99952	$-2.4 \times 10^{-4} \%$

the returned radius is 1 to ten digits, and the converged flux matches $\sin(Br)/Br$ to 3.2×10^{-6} (Figure 11).

The two approximations differ from each other far more than either differs from its own analytic solution, and the gap is entirely in the relaxation length: classical diffusion gives the larger radius throughout, by 1.1 % at $c = 1.02$ rising to 14.8 % at $c = 2$.

5.4 Verification

Three further checks, independent of the comparison in Table 3.

Convergence rate of the iteration. The error of source iteration falls by the dominance ratio k_1/k_0 per sweep. With the spherical modes $B_n = n\pi/r_{\text{out}}$ this is $(\Sigma_a + DB_1^2)/(\Sigma_a + DB_2^2)$, which at a *critical* radius collapses. Criticality means $\Sigma_a + DB_1^2 = \nu\Sigma_f$, and $B_2 = 2B_1$, so

$$\frac{k_1}{k_0} = \frac{\nu\Sigma_f}{4\nu\Sigma_f - 3\Sigma_a} = \frac{c}{4c - 3} \quad (30)$$

The diffusion coefficient has cancelled: *the convergence rate at criticality depends on c alone*. This is borne out exactly — the measured ratio agrees with (30) to six digits, and the two approximations take an identical number of sweeps at every c despite their different critical radii.

Table 4: Dominance ratio and sweep count to $|\Delta k| < 10^{-10}k$ at the critical radius. The sweep counts are identical for the classical and asymptotic approximations, as (30) requires.

c	$c/(4c - 3)$	measured ratio	sweeps
1.02	0.944444	0.944444	280
1.05	0.875000	0.875000	133
1.10	0.785714	0.785714	79
1.20	0.666667	0.666667	50
1.50	0.500000	0.500000	32
2.00	0.400000	0.400000	25

Neutron balance. For the converged critical flux, production must equal absorption plus leakage. The residual $|P - A - L|/P$ lies between 6×10^{-16} and 1.3×10^{-13} over the whole range, for both approximations. The informative term is the leakage: production and absorption are the same

volume integral times a constant, so their ratio $\nu\Sigma_f/(k\Sigma_a)$ holds identically whatever the flux. Leakage, by contrast, is computed from the boundary formula using only ϕ_{N-1} , and nothing forces it to close the balance unless the discrete operator conserves neutrons cell by cell and the boundary coefficient is exactly right. At $c = 1.5$ classical, $P = 32.2340$, $A = 21.4893$, $L = 10.7447$: about a third of the neutrons leak out, which is why the boundary treatment matters as much as it does below.

Boundary treatment. Table 5 exercises the second branch of (27) against a second, independent analytic route — $u \cot u = 1 - u/(B\ell_0)$ from `critical.dimensions.applied_bc`, which solves the condition on the flux shape instead of extrapolating. The numerical Robin radius reproduces it to 1.5×10^{-6} relative, the same discretisation error as the extrapolated case, confirming the boundary coefficient in both of its limits.

The two treatments are not interchangeable, though: they agree to first order in $B\ell_0$ and separate as the system shrinks, from 0.26 % apart at $c = 1.02$ to 4.5 % at $c = 2$. That gap is not solver error but the genuine ambiguity in what “the surface” means for a system a few mean free paths across.

Table 5: Extrapolated zero at $R + z_0$ against the condition $\phi + z_0\phi' = 0$ applied at R , classical diffusion, in mean free paths.

c	extrapolated	Robin	Robin analytic	Robin – extrapolated
1.02	12.158813	12.126930	12.126949	−0.26 %
1.10	5.069062	5.007865	5.007874	−1.21 %
1.50	1.898429	1.821650	1.821653	−4.04 %
2.00	1.147130	1.095638	1.095640	−4.49 %

6 Question 5: Bare Critical Mass of U-235 and Pu-239

Question 5 is Question 4 with the one-group benchmark cross sections of Sood, Forster & Parsons (2003), the mass following as $M_c = \rho \frac{4}{3}\pi R_c^3$. Both fissile materials have $\Sigma_t = 0.32640 \text{ cm}^{-1}$, a mean free path of 3.0637 cm, so both critical spheres are about two mean free paths in radius.

Table 6: Bare critical radius and mass from the $k = 1$ search on a 400-cell mesh. Numerical and analytic radii agree to four decimals in cm.

Material	Approximation	c	$\rho \text{ [g/cm}^3\text{]}$	$R_c \text{ [cm]}$	$\Sigma_t R_c$	$M_c \text{ [kg]}$
Pu-239	classical	1.50	15.7	5.8163	1.8984	12.940
Pu-239	asymptotic	1.50	15.7	5.1890	1.6937	9.188
U-235	classical	1.30	19.0	8.1031	2.6449	42.345
U-235	asymptotic	1.30	19.0	7.4339	2.4264	32.696

Because the mass goes as the cube of the radius, the moderate difference in relaxation length is strongly amplified: the classical radius exceeds the asymptotic one by 12.1 % (Pu-239) and 9.0 % (U-235), which becomes mass ratios of 1.41 and 1.30. Classical diffusion thus over-predicts the critical mass by 30–40 % here, and worsens with c . The asymptotic values are the ones to quote, since asymptotic diffusion with the exact $z_0(c)$ reproduces the exact-transport relation.

Question 4: critical radius from the $k = 1$ search vs. analytic

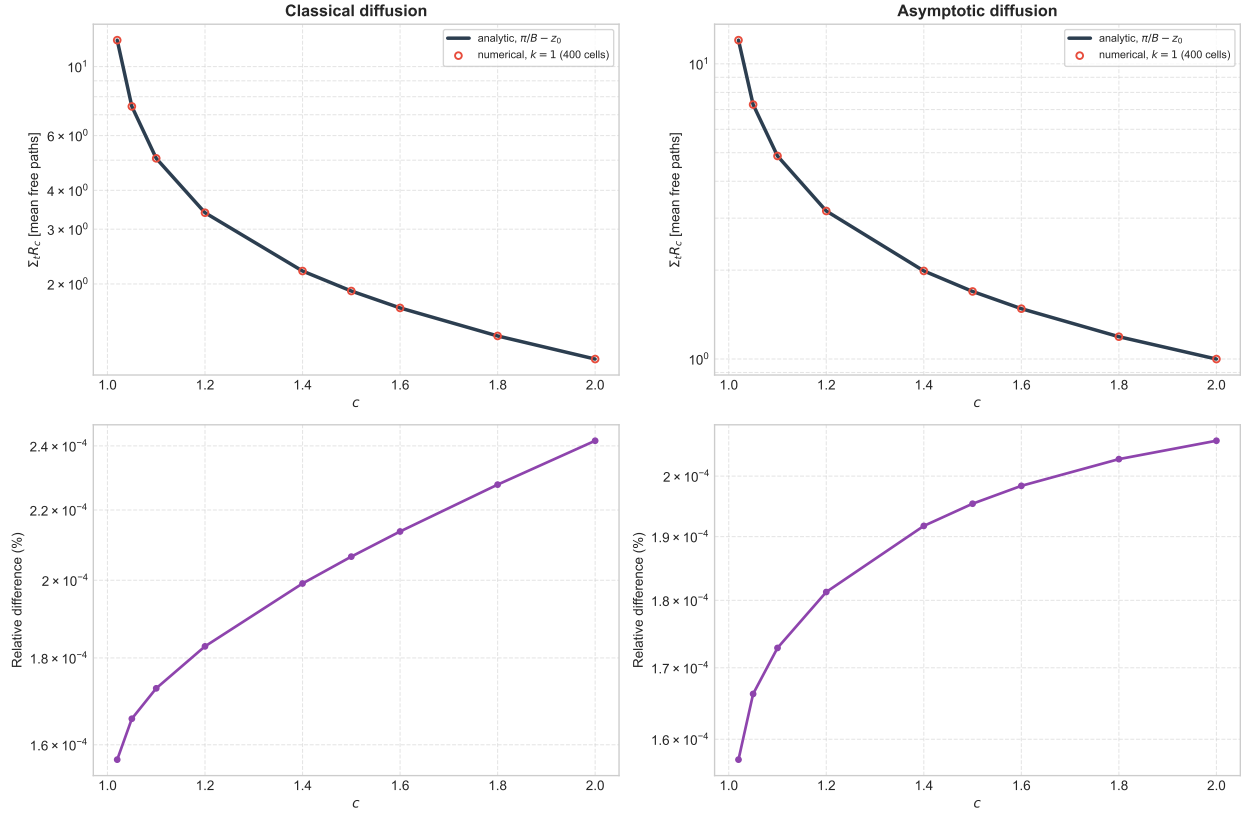


Figure 9: Critical radius from the $k = 1$ search (markers) against $\pi/B - z_0$ (solid), with the relative difference below.

Question 4: mesh convergence of the critical radius

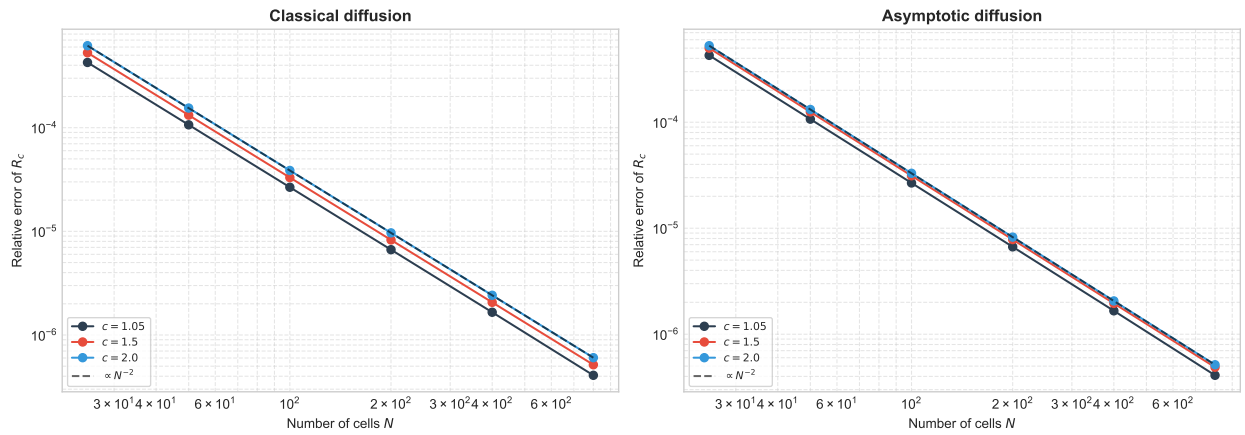


Figure 10: Mesh convergence of the critical radius, following the second-order guide $\propto N^{-2}$ at every refinement.

Question 4: critical flux shape vs. the fundamental mode

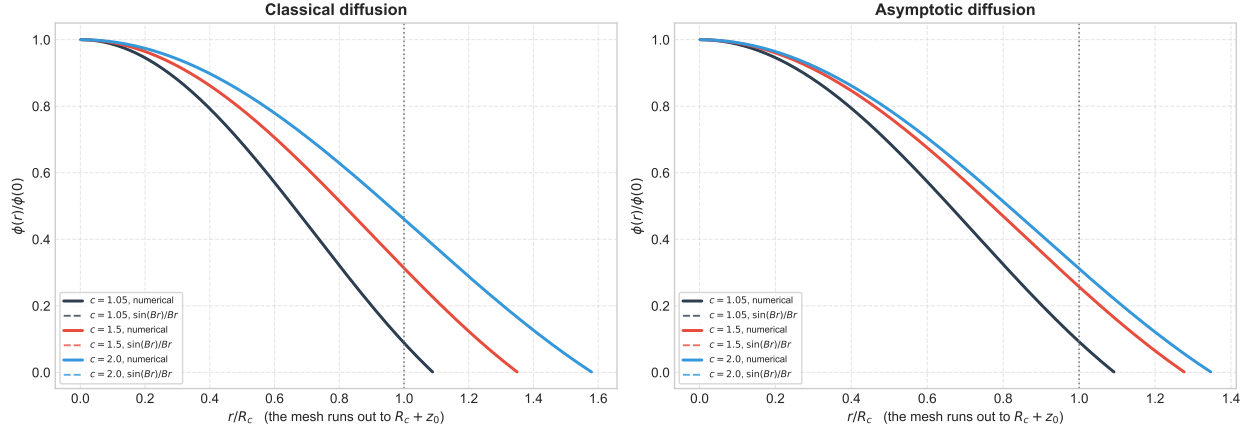


Figure 11: Converged critical flux (solid) against the fundamental mode $\sin(Br)/Br$ (dashed), normalised at the centre. The mesh continues past $r/R_c = 1$ to the extrapolated radius, where the flux is driven to zero.

6.1 Sensitivity to the Boundary Treatment

The masses above use the extrapolated zero. Applying the same condition directly at the physical surface instead moves them by far more than any numerical error, again because the mass goes as the cube of a radius only a couple of mean free paths long.

Table 7: Critical mass under the two boundary treatments. Both reproduce their own analytic result to 1.5×10^{-6} , so the spread is model uncertainty, not solver error.

Material	Approximation	M_c extrapolated [kg]	M_c Robin [kg]	difference
Pu-239	classical	12.940	11.432	−11.7 %
Pu-239	asymptotic	9.188	8.307	−9.6 %
U-235	classical	42.345	38.679	−8.7 %
U-235	asymptotic	32.696	30.212	−7.6 %

Both treatments reproduce their own analytic result, so this 8–12 % spread is model uncertainty rather than solver error, and it is comparable to the difference between the two diffusion approximations themselves. Any single quoted mass should be read with it in mind.

These are one-group, isotropic-scattering results, not predictions of real critical masses: at about two mean free paths the boundary layer is a large fraction of the system — roughly a third of the neutrons leak out — so diffusion of either flavour is well outside its regime, and the transport benchmark radii of Sood *et al.* are the appropriate reference. The remaining three benchmark rows (H_2O , Fe, Na) have $c \leq 1$ and are non-multiplying, so no bare critical sphere exists for them at any radius.

A note on the U-235 data. Two rows were in circulation. The assignment table ($\Sigma_c = 0.013056$, $\Sigma_s = 0.248064$, $\Sigma_t = 0.32640$) implies $c = 1.3000$, matching its quoted value, and is what is used above. The alternative row ($\Sigma_c = 0.015672$, $\Sigma_s = 0.180448$, $\Sigma_t = 0.26140$) is quoted with $c = 1.50$ but implies $c = 1.3646$, so it is not self-consistent; through the same code it gives $R_c = 8.9412$

cm, $M_c = 56.89$ kg classically and 8.1161 cm, 42.55 kg asymptotically — roughly 30 % more mass. Pu-239 is identical in both sources.

Question 5: criticality of the bare benchmark spheres

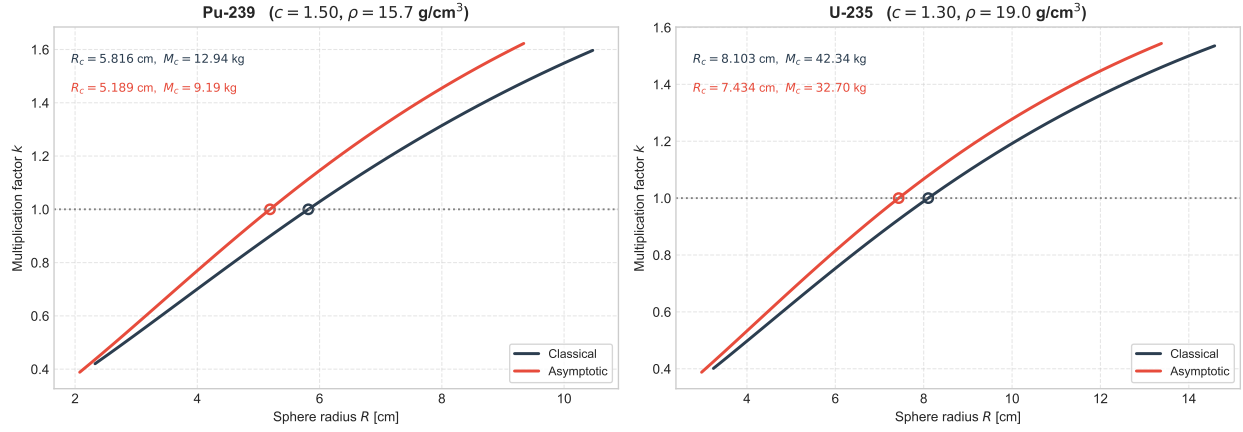


Figure 12: Multiplication factor against sphere radius for the two fissile materials, both approximations. The circles mark the $k = 1$ crossing.